

"Falcon 2.0T: Stretched timing chain sets crankshaft/camshaft correlation fault"

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Area: bulletins

1 Condition

Some Falcon 2.0T vehicles may set DTC P0016 "Crankshaft Position - Camshaft Position Correlation (Bank 1 Sensor A)" with the malfunction indicator illuminated. Customers may report a rattle on cold start, hard starting, and a noticeable loss of power. The PCM detects that the measured camshaft phase no longer agrees with the crankshaft reference, which triggers P0016 and may force the engine into a reduced-power mode. See Falcon Engine Service Manual :: Timing System for the timing reference architecture associated with this fault.

2 Cause

The root cause is mechanical stretch of the primary timing chain, part number TC-3340, which allows the camshaft to retard relative to the crankshaft beyond the correlation threshold. Chain stretch is accelerated by degraded or contaminated engine oil and extended drain intervals, so confirm oil condition and service history per PROC-OIL before any teardown. Detailed wear-limit measurement for TC-3340 is documented in Falcon Engine Service Manual :: Timing Chain Service .

3 Repair Procedure

Verify crankshaft and camshaft position sensor signals first by running PROC-CPS-CMP; if the camshaft position sensor CMP-3382 is reading correctly but correlation is still out of range, the timing chain is the likely fault. Replace timing chain TC-3340 following Falcon Engine Service Manual :: Timing Chain Service , and if CMP-3382 is found damaged or contaminated during the repair, replace it per Falcon Engine Service Manual :: Camshaft Position Sensor Replacement . Complete an oil and filter service per PROC-OIL using the specified viscosity, then clear codes and confirm P0016 does not return. Additional sensor reference is available in Falcon Engine Service Manual :: Position Sensors and the master Falcon Engine Service Manual .